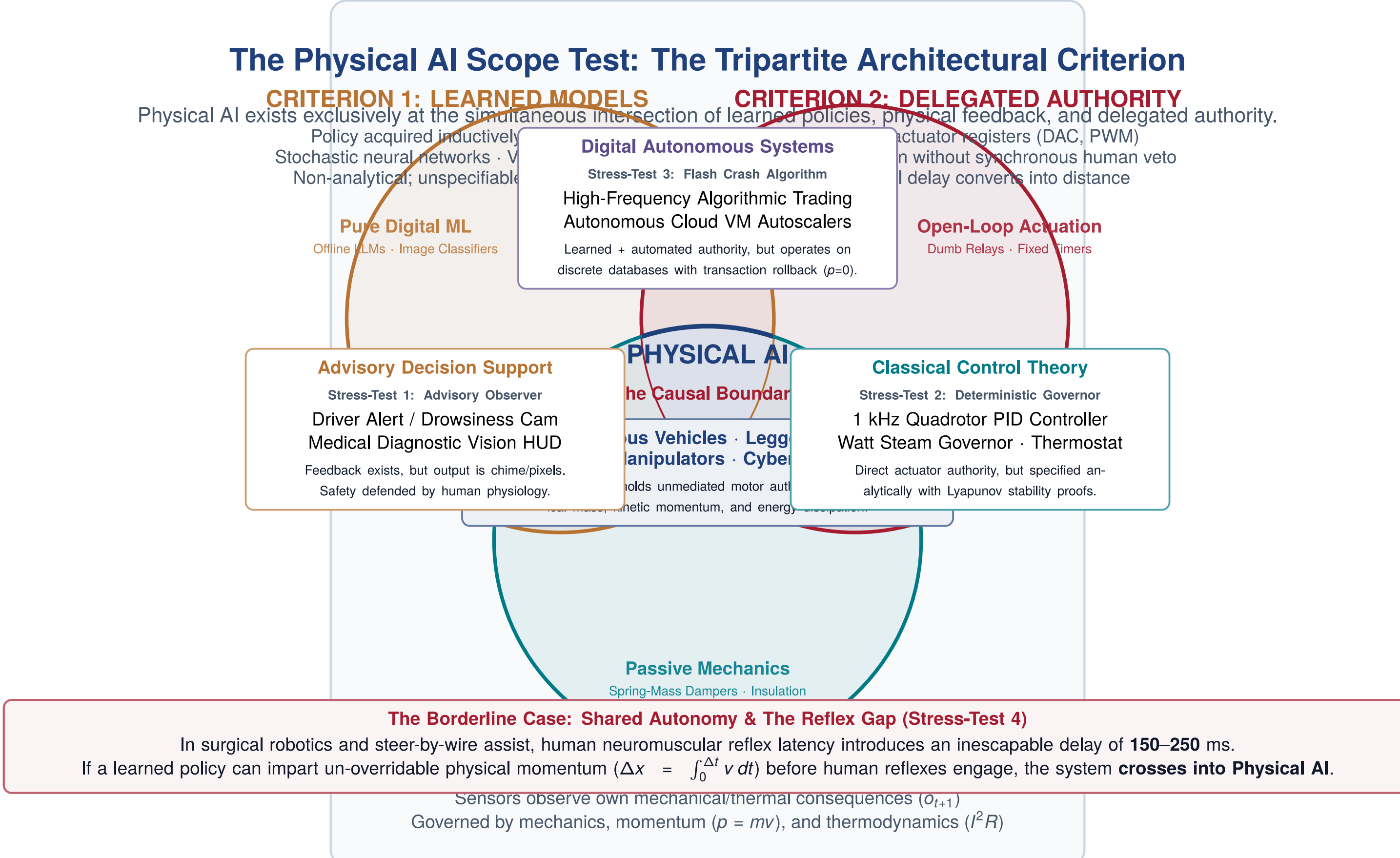
The Physical AI Scope Test: The Tripartite Architectural Criterion
CRITERION 1: LEARNED MODELS
CRITERION 2: DELEGATED AUTHORITY
Physical AI exists exclusively at the simultaneous intersection of learned policies, physical feedback, and delegated authority.
Policy acquired inductively
actuator registers (DAC, PWM)
Stochastic neural networks
without synchronous human veto
Non-analytical; unspecifiable
delay converts into distance
Digital Autonomous Systems
Stress-Test 3: Flash Crash Algorithm
High-Frequency Algorithmic Trading
Autonomous Cloud VM Autoscalers
Learned + automated authority, but operates on discrete databases with transaction rollback (p=0).
Pure Digital ML
Offline LLMs · Image Classifiers
Open-Loop Actuation
Dumb Relays · Fixed Timers
PHYSICAL AI
Advisory Decision Support
Stress-Test 1: Advisory Observer
Driver Alert / Drowsiness Cam
Medical Diagnostic Vision HUD
Feedback exists, but output is chime/pixels.
Safety defended by human physiology.
Classical Control Theory
Stress-Test 2: Deterministic Governor
1 kHz Quadrotor PID Controller
Watt Steam Governor · Thermostat
Direct actuator authority, but specified analytically with Lyapunov stability proofs.
Passive Mechanics
Spring-Mass Dampers · Insulation
The Borderline Case: Shared Autonomy & The Reflex Gap (Stress-Test 4)
In surgical robotics and steer-by-wire assist, human neuromuscular reflex latency introduces an inescapable delay of **150–250** ms.
If a learned policy can impart un-overridable physical momentum ($\Delta x = \int_0^{\Delta t} v\, dt$) before human reflexes engage, the system **crosses into Physical AI**.
Sensors observe own mechanical/thermal consequences (o_{t+1})
Governed by mechanics, momentum ($p = mv$), and thermodynamics (I^2R)